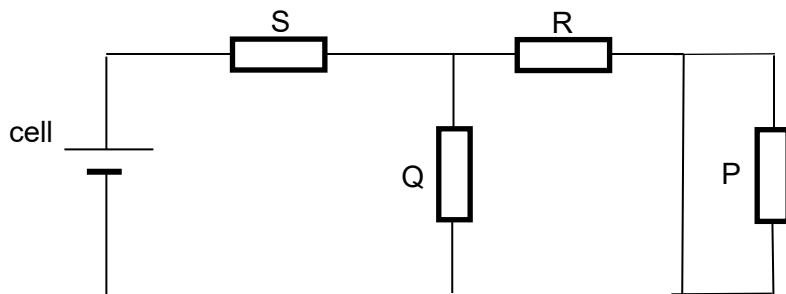


- 23 A cell is connected across four identical resistors P, Q, R and S. If the source is supplying a total power of 12.0 W, what is the power dissipated heat in resistor R?



- A** 2.0 W **B** 3.0 W **C** 4.0 W **D** 5.0 W

Ans: A

No current will flow through P.

Let the resistance of each resistor be r and e.m.f. of source be $\mathcal{E}$.

Q and R are connected in parallel, so their effective resistance is $0.5r$

Total resistance of circuit is $1.5r$

$$\text{Total power } P = \frac{\mathcal{E}^2}{1.5r} = 12.0 \text{ W}$$

$$\text{By potential divider principle, voltage across R} = \frac{0.5r}{1.5r} \times \mathcal{E} = \frac{\mathcal{E}}{3}$$

$$\text{Power dissipated across resistor R, } P = \frac{\left(\frac{\mathcal{E}}{3}\right)^2}{r} = \frac{\mathcal{E}^2}{9r} = 2.0 \text{ W}$$